Omni-Compass benchmark: Kubernetes native vs Kubernetes + Omni-Compass

Two identical kind clusters (1 control plane + 6 workers), same add-ons, workload (php-apache + HPA, target 50), load schedule and capture. **Native**: Omni-Compass not running. **Omni**: Omni-Compass drives the HPA target, the node pool (cordon/drain/uncordon), the power cap (in-place CPU limits on the app pods, enforced by the kernel), heat (harness law on live power), security (hold signal) and the rollout guard. Response times are real HTTP requests timed every 5 s. Power is a declared model (idle 100 W + 150 W x utilisation per worker in service; parked workers on standby at idle power), not a meter.

Gauges

Gauge	Native	Omni-Compass	Change
duration (min)	20.03	20.23	
worker nodes in service, mean	6.00	6.00	+0.0%
worker nodes in service, min	6.00	6.00	+0.0%
worker nodes in service, max	6.00	6.00	+0.0%
node-hours	2.00	2.02	+1.0%
power (W), mean	627	640	+2.1%
power (W), peak	650	686	+5.5%
energy (Wh)	209	216	+3.1%
energy, parked workers still on at idle power (Wh)	209	216	+3.1%
CPU used (cores), mean	0.833	1.18	+42.2%
CPU allocatable (cores), mean	24.00	24.00	+0.0%
utilisation (used / allocatable)	0.035	0.049	+42.2%
energy per core-hour (Wh)	753	541	-28.2%
node-hours per core-hour	7.21	5.07	-29.7%
pending pods, pod-minutes	0.000	0.533	new
pending pods, peak	0.000	2.00	new
HPA replicas, mean	8.60	9.26	+7.8%
HPA replicas, peak	10.00	10.00	+0.0%
HPA shortfall (desired > current), minutes	1.32	0.517	-60.8%
requests timed	1,100	1,105	+0.5%
time over the response line (% of samples)	0.000	0.000	0
response time (ms), mean	94.81	88.30	-6.9%
response time (ms), median	84.45	74.70	-11.5%
response time (ms), 95th percentile	211	142	-32.8%
response time (ms), 99th percentile	300	170	-43.2%
failed requests (%)	0.000	0.000	0

Significance (2-minute blocks: native 10, omni 10; bootstrap 95% interval)

Metric	Native	Omni-Compass	Change	95% CI of difference	Verdict
node_hours_per_core_hour	7.9279	5.8445	-26.3%	[-3.9184, -0.2051]	GAIN
kwh_per_core_hour	0.8247	0.6178	-25.1%	[-0.3844, -0.0237]	GAIN
utilisation	0.0343	0.0485	+41.5%	[+0.0020, +0.0278]	GAIN
pending_pod_minutes_per_hour	0.0000	1.5238	n/a	[+0.0000, +4.5714]	STEADY, no significant difference
hpa_shortfall_minutes_per_hour	3.7917	1.4762	-61.1%	[-7.6608, +1.5599]	STEADY, no significant difference

What Omni-Compass did

• decisions: 120 (one per minute)

• HPA target writes: 0

• node-pool resizes: 3

• scheduling-floor adds (pending pods): 0

• power-cap pod resizes (in place): 0

• rollout actions: 4

• power cap decided per minute: 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 0.695 0.695 0.695 1.0 0.695 1.0 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695

Reset

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restored target: 50
workers in service: 6 of 6
Omni records left after kill: none
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Minute by minute

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
0	6.00	6.00	616	615	4.25	5.00	0.000	0.000	0.545	0.509
1	6.00	6.00	617	624	5.00	6.00	0.000	0.000	0.531	0.786
2	6.00	6.00	615	621	5.00	7.00	0.000	0.000	0.497	0.687
3	6.00	6.00	619	633	5.00	7.75	0.000	0.500	0.600	0.985
4	6.00	6.00	629	648	6.75	10.00	0.000	0.000	0.870	1.39
5	6.00	6.00	628	652	8.00	10.00	0.000	0.000	0.891	1.44
6	6.00	6.00	633	642	9.00	10.00	0.000	0.000	1.00	1.22
7	6.00	6.00	646	671	9.67	10.00	0.000	0.000	1.35	2.04
8	6.00	6.00	642	671	10.00	10.00	0.000	0.000	1.27	2.03
9	6.00	6.00	645	676	10.00	10.00	0.000	0.000	1.32	2.15
10	6.00	6.00	642	672	10.00	10.00	0.000	0.000	1.22	2.03
11	6.00	6.00	618	622	10.00	10.00	0.000	0.000	0.595	0.668
12	6.00	6.00	615	624	10.00	10.00	0.000	0.000	0.520	0.729
13	6.00	6.00	624	631	10.00	10.00	0.000	0.000	0.705	0.947
14	6.00	6.00	632	646	10.00	10.00	0.000	0.000	0.962	1.34
15	6.00	6.00	633	644	10.00	10.00	0.000	0.000	0.980	1.31
16	6.00	6.00	629	649	10.00	10.00	0.000	0.000	0.905	1.37
17	6.00	6.00	621	631	10.00	10.00	0.000	0.000	0.754	0.948
18	6.00	6.00	617	622	10.00	10.00	0.000	0.000	0.561	0.705
19	6.00	6.00	619	621	10.00	10.00	0.000	0.000	0.586	0.682

Limits

- kind nodes are containers sharing one CI machine; timings and CPU are noisier than real servers.
- The two arms ran on two different CI machines at the same time; machine-to-machine variation is part of the noise.
- Energy is modelled from utilisation and nodes in service with the declared constants above.